Ex No. 14
Date : 08/10/24

Packet Sniffing Using Raw Sockets

AIM :-

To write a program using a raw socket to capture network packets & display the source MAC address, protocol number & packet length.

Algorithm :-

1. Start the Program
2. Import the socket and struct Module
3. Create a Raw Socket using AF-Packet for packet-level access. socket-RAW for receiving raw packets.
4. Display a message indicating that the packet sniffer has started.
5. Continously receive packets using recvfrom (65535).
6. Extract : destination MAC address source MAC address ethernet Protocol type.
7. Convert the MAC address from bytes into readable format.
8. Display source & destination MAC Protocol.
9. Stop the program.

Program :-

```
import socket
import struct
raw_socket = socket.socket(socket.AF_PACKET,
socket.SOCK_RAW, socket.ntohs(3))
Print("Packet Sniffer Started....")
Print("Waiting for Packets ....\n")
```

Output :-

Ether / IP / UDP 172.16.79.235. 44024 > 239.0.0.8. 51
Source IP : 172.16.79.235
Destination IP : 239.0.0.8
Protocol : 17
Packet Length : 1078

Ether / IP / UDP 172.16.79.235.44024 > 239.0.0.8.5210
Source IP : ~~172.16.79~~ 239.0.0.8
Protocol : 17
Packet Length : 1078.

Ether / IP / UDP 172.16.79.8 235.44024 > 239.0.0.8.52
Source IP : 239.0.0.8
Protocol : 17
Packet Length : 1078.

```
while True :
    Packet . address = raw_socket . recv from (65535)
    ethernet_header = Packet [ : 14]
    dest_mac , src_mac , protocol = struct . unpack
    ( ethernet_header)
    dest_mac = ":" join ( "%02x" Byte for Byte in
    src_mac)
    Print ("Source mac :", src_mac)
    Print ("Destination mac :", dest_mac)
    Print ("Protocol :", Protocol)
    Print ("Packet Length :", len(Packet))
```

Result :-

The RAW Socket Packet - sniffing program was successfully executed.